From the above equation it follows that a phase variation of the type

$$\exp\left[i\frac{k}{2R}(x^2+y^2)\right]$$

(on the x-y plane) represents a *diverging* spherical wave of radius R. If we now compare the above expression with Eqs. (20.36) and (20.37) we can say that the radius of curvature of the phase front of the propagating Gaussian beam is given by Eq. (20.37). Thus as the beam propagates, the phase front which was plane at $z = 0$ becomes curved. In Sec. 27.5, we will use the above analysis to determine conditions for a Gaussian beam to resonate between two spherical mirrors. Here, as a simple application, we will consider a resonator configuration consisting of a plane mirror and a spherical mirror separated by a distance d (see Fig. 20.10); indeed such a configuration is used to produce a single transverse mode oscillation in a ruby laser. On the plane $z = 0$, the phase front is plane and for the beam to resonate, the phase front must have a radius of curvature equal to the radius of curvature of the mirror:

$$R = d\left[1+\frac{\pi^2 w_0^4}{\lambda^2 d^2}\right] \quad \Rightarrow \quad w_0^2 = \frac{\lambda d}{\pi}\sqrt{\left(\frac{R}{d}-1\right)} \qquad \textbf{(20.41)}$$

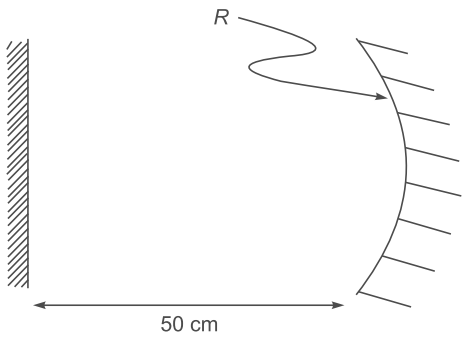


Fig. 20.10 A simple resonator consisting of a plane mirror and a concave mirror of radius R.

Example 20.3 We assume $\lambda \approx 0.6328\ \mu$m, $d \approx 50$ cm and $R \approx 100$ cm; which are typical parameters for a He-Ne laser. Simple calculations will show that $w_0 \approx 0.32$ mm. If we increase R to 200 cm, we will get $w_0 \approx 0.38$ mm. Obviously R should be greater than d. In Sec. 27.5, we will consider the more general case of a Gaussian beam resonating between two mirrors of radii of curvatures R_1 and R_2 as shown in Fig. 27.26.

20.6 DIFFRACTION AT A STRAIGHT EDGE LO 4

Before we discuss the straight edge diffraction pattern, we introduce the Fresnel integrals.

Fresnel Integrals: Fresnel integrals are defined by the following equations:

$$C(\tau) = \int_0^{\tau} \cos\left(\frac{1}{2}\pi u^2\right) du \qquad \textbf{(20.42)}$$

and

$$S(\tau) = \int_0^{\tau} \sin\left(\frac{1}{2}\pi u^2\right) du \qquad \textbf{(20.43)}$$

Since the integrands are even functions of τ, the Fresnel integrals $C(\tau)$ and $S(\tau)$ are odd functions of τ:

$$C(-\tau) = -C(\tau) \quad \text{and} \quad S(-\tau) = -S(\tau) \qquad \textbf{(20.44)}$$

Further, since

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\, dx = \sqrt{\frac{\pi}{\alpha}} \qquad \textbf{(20.45)}$$

we have

$$\int_{-\infty}^{\infty} e^{i\pi u^2/2}\, du = \sqrt{\frac{\pi}{-i\pi/2}} = \sqrt{2}\, e^{i\pi/4} = (1+i) \qquad \textbf{(20.46)}$$

Now,

$$\int_{-\infty}^{+\infty} \exp\left[i\frac{\pi u^2}{2}\right] du$$

$$= 2\left[\int_0^{\infty} \cos\left(\frac{1}{2}\pi u^2\right) du + i\int_0^{\infty} \sin\left(\frac{1}{2}\pi u^2\right) du\right]$$

$$= 2\,[C(\infty) + i\,S(\infty)]$$

Thus, using Eq. (20.46), we get $C(\infty) = \frac{1}{2} = S(\infty)$.

To summarize, the Fresnel integrals have the following important properties:

$$C(\infty) = S(\infty) = \frac{1}{2};\ C(0) = S(0) = 0 \qquad \textbf{(20.47)}$$

$$C(-\tau) = -C(\tau) \quad \text{and} \quad S(-\tau) = -S(\tau) \qquad \textbf{(20.48)}$$

The values of the Fresnel integrals for typical values of τ are tabulated in Table 20.1.

Table 20.1 Table of Fresnel Integrals*

$$C(\tau)=\int_0^{\tau}\cos\left(\frac{\pi}{2}v^2\right)dv;\quad S(\tau)=\int_0^{\tau}\sin\left(\frac{\pi}{2}v^2\right)dv$$

τ	$C(\tau)$	$S(\tau)$	τ	$C(\tau)$	$S(\tau)$
0.0	0.00000	0.00000	2.6	0.38894	0.54999
0.2	0.19992	0.00419	2.8	0.46749	0.39153
0.4	0.39748	0.03336	3.0	0.60572	0.49631
0.6	0.58110	0.11054	3.2	0.46632	0.59335
0.8	0.72284	0.24934	3.4	0.43849	0.42965
1.0	0.77989	0.43826	3.6	0.58795	0.49231
1.2	0.71544	0.62340	3.8	0.44809	0.56562
1.4	0.54310	0.71353	4.0	0.49843	0.42052
1.6	0.36546	0.63889	4.2	0.54172	0.56320
1.8	0.33363	0.45094	4.4	0.43833	0.46227
2.0	0.48825	0.34342	4.6	0.56724	0.51619
2.2	0.63629	0.45570	4.8	0.43380	0.49675
2.4	0.55496	0.61969	5.0	0.56363	0.49919
			∞	0.5	0.5

Figure 20.11 gives a parametric representation of the Fresnel integrals and is known as the **Cornu's spiral**. The horizontal and the vertical axes represent $C(\tau)$ and $S(\tau)$, respectively and the numbers written on the spiral are the values of τ. For example, as can be seen from the figure, for $\tau = 1.0$, $C(\tau) \approx 0.77989$ and $S(\tau) \approx 0.43826$.

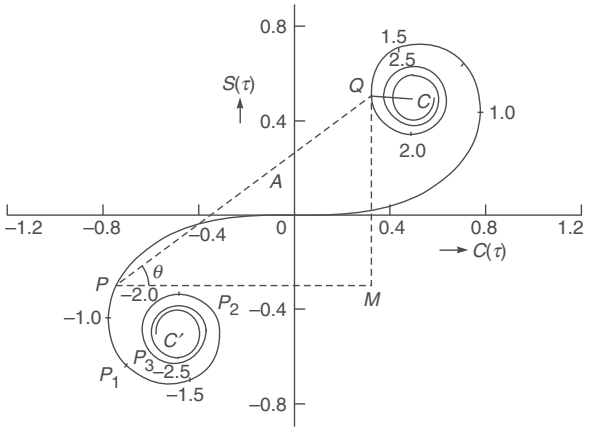


Fig. 20.11 The Cornu's spiral which is a parametric plot of $C(\tau)$ and $S(\tau)$.

We will now use Eq. (20.23) to calculate the diffraction pattern of a plane wave incident normally on a straight edge (see Fig. 20.12). Obviously, there will be no variation of intensity along the x-axis and, therefore, without any loss of generality, we may assume the coordinates of an arbitrary point P (on the screen) to be $(0, y)$, where the origin has been assumed to be on the edge of the geometrical shadow. If the x and y coordinates of an arbitrary point M on the plane of the straight edge are denoted by ξ and η, then

$$\begin{aligned} r = MP &= [\xi^2 + (\eta - y)^2 + d^2]^{1/2} \\ &= d\left[1 + \frac{\xi^2 + (\eta - y)^2}{d^2}\right]^{1/2} \\ &\approx d + \frac{\xi^2 + (\eta - y)^2}{2d} \end{aligned} \qquad \textbf{(20.49)}$$

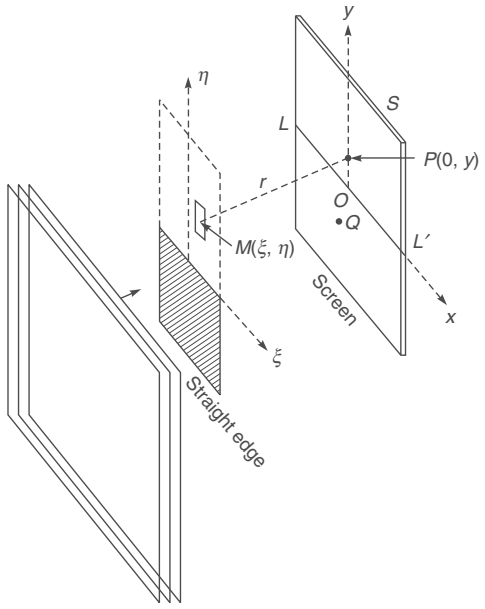


Fig. 20.12 Diffraction of a plane wave incident normally on a straight edge.

where d is the distance between the straight edge and the screen. On substituting the expression for r from Eq. (20.49) in Eq. (20.21), we obtain

$$u(P) \approx -\frac{i}{\lambda}\frac{A}{d}\int_{-\infty}^{\infty} d\xi \int_0^{\infty} d\eta \exp\left[ik\left\{d + \frac{\xi^2 + (\eta - y)^2}{2d}\right\}\right] \qquad \textbf{(20.50)}$$

where, in the denominator of the integrand, we have replaced r by its minimum value**, d. In order to express the above expression in terms of the Fresnel integrals, we introduce two dimensionless variables u and v such that

* Table adapted from Ref. 20.5; a more detailed table (with greater accuracy) has been given there.

** This is justified because in carrying out the integration, only a small region around the point $r = d$ contributes; the contribution due to far-off points is small because of the rapid oscillations of the exponential term in the integrand (see also footnote in Sec. 19.2).

$$\text{and} \quad \left.\begin{aligned} \frac{1}{2}\pi u^2 &= \frac{k}{2d}\xi^2 = \frac{\pi}{\lambda d}\xi^2 \\ \frac{1}{2}\pi v^2 &= \frac{k}{2d}(\eta - y)^2 = \frac{\pi}{\lambda d}(\eta - y)^2 \end{aligned}\right\}$$

Thus we may assume u and v to be *defined* by the following equations:

$$\text{and} \quad \left.\begin{aligned} u &= \sqrt{\frac{2}{\lambda d}}\,\xi \\ v &= \sqrt{\frac{2}{\lambda d}}\,(\eta - y) \end{aligned}\right\} \qquad \textbf{(20.51)}$$

With these substitutions, Eq. (20.51) becomes

$$u(P) = -\frac{i}{2}u_0 \int_{-\infty}^{+\infty} \exp\left(\frac{i\pi u^2}{2}\right) du \int_{v_0}^{\infty} \exp\left(\frac{i\pi v^2}{2}\right) dv \qquad \textbf{(20.52)}$$

where

$$v_0 = -\sqrt{\frac{2}{\lambda d}}\,y \qquad \textbf{(20.53)}$$

and

$$u_0 = A\,e^{ikd}$$

represents the field at the point P in the absence of the straight edge. In order to calculate the intensity distribution we use the Fresnel integrals; thus,

$$\int_{-\infty}^{+\infty} \exp\left[i\frac{\pi u^2}{2}\right] du = 2\,[C(\infty) + S(\infty)]$$

$$= (1 + i) \qquad \textbf{(20.54)}$$

Further,

$$\int_{v_0}^{\infty} \exp\left(\frac{i\pi v^2}{2}\right) dv = \left[\int_0^{\infty} \cos\left(\frac{\pi}{2}v^2\right) dv - \int_0^{v_0} \cos\left(\frac{\pi}{2}v^2\right) dv\right]$$

$$+ i\left[\int_0^{\infty} \sin\left(\frac{\pi}{2}v^2\right) dv - \int_0^{v_0} \sin\left(\frac{\pi}{2}v^2\right) dv\right]$$

$$= \left[\frac{1}{2} - C(v_0)\right] + i\left[\frac{1}{2} - S(v_0)\right] \qquad \textbf{(20.55)}$$

Substituting in Eq. (20.52), we obtain

$$u(P) = -\frac{i}{2}u_0(1+i)\left[\left\{\frac{1}{2} - C(v_0)\right\} + i\left\{\frac{1}{2} - S(v_0)\right\}\right]$$

$$= \frac{1-i}{2}u_0\left[\left\{\frac{1}{2} - C(v_0)\right\} + i\left\{\frac{1}{2} - S(v_0)\right\}\right] \qquad \textbf{(20.56)}$$

It is of interest to note that a large value of y corresponds to a point which is very far above the edge of the geometrical shadow. For such a point v_0 would tend to $-\infty$ [see Eq. (20.53)] and we would obtain

$$u(P) = \frac{1-i}{2}u_0\left[\left(\frac{1}{2} + \frac{1}{2}\right) + i\left(\frac{1}{2} + \frac{1}{2}\right)\right]$$

$$= u_0 \qquad \textbf{(20.57)}$$

Thus, as expected, the amplitude at such a point is the same as that in the absence of the edge. This also justifies the value of the constant given by Eq. (19.3). On the other hand, when the point P is deep inside the geometrical shadow (i.e., when $y \to -\infty$ and hence $v_0 \to \infty$), we obtain

$$C(v_0) = S(v_0) \to \tfrac{1}{2}$$

giving

$$u(P) \to 0$$

as it should indeed be. The intensity distribution corresponding to Eq. (20.56) would be given by

$$I(P) = \frac{1}{2}I_0\left[\left\{\frac{1}{2} - C(v_0)\right\}^2 + \left\{\frac{1}{2} - S(v_0)\right\}^2\right] \qquad \textbf{(20.58)}$$

If the point P is such that it lies on the edge of the geometrical shadow (i.e., on the line LL' (see Fig. 20.12) then $y = 0$ and hence $v_0 = 0$; thus

$$I(P) = \frac{1}{2}I_0\left[\frac{1}{4} + \frac{1}{4}\right] = \frac{1}{4}I_0 \qquad \textbf{(20.59)}$$

where we have used the fact that $C(0) = S(0) = 0$. Thus the intensity on the edge of the geometrical shadow is $1/4^{\text{th}}$ of the intensity that would have been in the absence of the edge. In order to determine the field at an arbitrary point P, we may use Table 20.1 to calculate the RHS of Eq. (20.58). The intensity variation is plotted in Fig. 20.16 from which one can make the following observations:

(i) Figure 20.13 represents a universal curve, i.e., for given values of λ and d, one simply has to calculate v_0 as the observation point moves along the y-axis. For example, the first three maxima occur at

$$\text{and} \quad \left.\begin{aligned} v_0 &\approx -1.22 \ \text{ with } \ I \approx 1.37\,I_0 \\ v_0 &\approx -2.34 \ \text{ with } \ I \approx 1.20\,I_0 \\ v_0 &\approx -3.08 \ \text{ with } \ I \approx 1.15\,I_0 \end{aligned}\right\} \text{ maxima}$$

Similarly, the first three minima occur at

$$\text{and} \quad \left.\begin{aligned} v_0 &\approx -1.87 \ \text{ with } \ I \approx 0.778\,I_0 \\ v_0 &\approx -2.74 \ \text{ with } \ I \approx 0.843\,I_0 \\ v_0 &\approx -3.39 \ \text{ with } \ I \approx 0.872\,I_0 \end{aligned}\right\} \text{ minima}$$

Thus, as we go inside the geometrical shadow, the intensity modulation decreases (see Fig. 20.13).

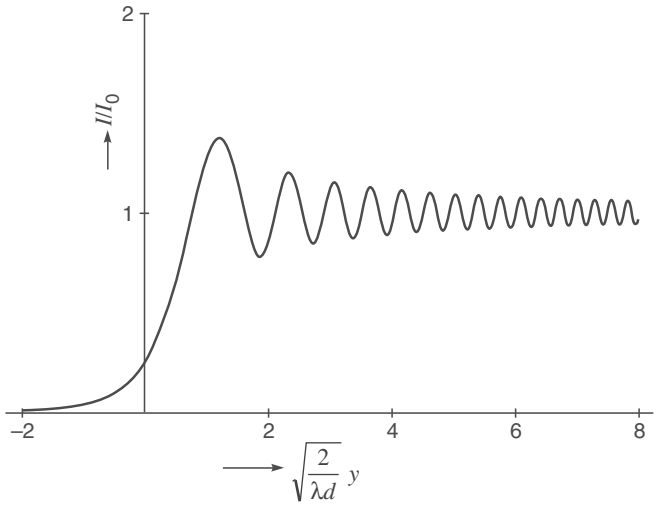


Fig. 20.13 The intensity variation corresponding to the straight edge diffraction pattern.

(ii) For a given experimental set-up, the determination of the positions of maxima and minima is quite straightforward. For example, for $\lambda = 6 \times 10^{-5}$ cm and $d = 120$ cm

$$y = -\sqrt{\frac{\lambda d}{2}}\, v_0 = -0.06\, v_0 \text{ cm}$$

Thus, the first three maxima will occur at

$$y \approx 0.732,\ 1.404 \quad \text{and} \quad 1.848 \text{ mm}$$

respectively. Similarly, the first three minima will occur at

$$y \approx 1.122,\ 1.644 \text{ and } 2.034 \text{ mm}$$

respectively.

(iii) As we go inside the geometrical shadow the intensity monotonically decreases to zero.

(iv) One could have also studied the intensity variation directly from the Cornu's spiral (see Fig. 20.11). This is due to the fact that associated with the Cornu's spiral, we have the following interesting property: let us write

$$[C(\tau_2) - C(\tau_1)] + i[S(\tau_2) - S(\tau_1)] \equiv A\, e^{i\theta} \qquad \textbf{(20.60)}$$

Thus,

$$[C(\tau_2) - C(\tau_1)] = A \cos\theta$$

and

$$[S(\tau_2) - S(\tau_1)] = A \sin\theta$$

Let the points P and Q on the Cornu's spiral (see Fig. 20.11) correspond to $\tau = \tau_1$ and $\tau = \tau_2$ respectively. It is obvious that

$$PM = [C(\tau_2) - C(\tau_1)] = A \cos\theta$$

and

$$QM = [S(\tau_2) - S(\tau_1)] = A \sin\theta$$

Thus the length of the line joining the points P and Q will be A and the angle that the line makes with the abscissa will be θ. In order to use the Cornu's spiral we rewrite Eq. (20.56):

$$u = \frac{1-i}{2}\, u_0 \left[\left\{ \frac{1}{2} - C(v_0) \right\} + i \left\{ \frac{1}{2} - S(v_0) \right\} \right]$$

Let us first consider a point of observation Q in the geometrical shadow region. Consequently v_0 will be positive. Let the point Q on the spiral (see Fig. 20.11) correspond to $\tau = v_0$. Since the point C in the curve corresponds to $\tau = \infty$, we have

$$\left\{ \frac{1}{2} - C(v_0) \right\} + i \left\{ \frac{1}{2} - S(v_0) \right\} = (QC)\, e^{i\psi}$$

where ψ is the angle that QC makes with the abscissa [see Eq. (20.60)]. Thus,

$$u(Q) = \frac{1-i}{2}\, (QC)\, e^{i\psi}\, u_0$$

or

$$I(Q) = \frac{1}{2}\, (QC)^2\, I_0 \qquad \textbf{(20.61)}$$

We can easily see that as the point of observation moves into the shadow region, the value of v_0 increases. Thus the point Q keeps on moving on the spiral towards the point C and the length QC decreases uniformly. Hence in the shadow region the intensity uniformly decreases to zero (see Figs. 20.13 and 20.14).

As we move away from the edge of the geometrical shadow to the illuminated region, the value of v_0 becomes negative and the corresponding point P (on the Cornu's spiral) lies in the third quadrant as shown in Fig. 20.11. The intensity is again given by

$$I(P) = \frac{1}{2}\, (PC)^2\, I_0$$

As the value of v_0 becomes more and more negative, the length PC keeps on increasing till the point P reaches P_1 which corresponds to $v_0 \approx -1.22$. The intensity at this point is maximum and the length $P_1C \approx 1.66$. Thus, the corresponding intensity is

$$I(P_1) \approx \frac{1}{2}\, (1.66)^2\, I_0 \approx 1.37\, I_0 \qquad \textbf{(20.62)}$$

As the value of v_0 becomes further negative the length PC starts decreasing till it reaches the point P_2. Thus,